Markov Sail Duality from Christoffel matrix paths

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Task 1. Prove the Markov Sail Duality conjecture of Evans, Veselov, and Winn [1, Conjecture 8.3]. We prove an explicit formula for every interior coefficient on the sail segments involved in that conjecture. Throughout, $2 \le a < b$ are coprime, $P = P_{a/b}(u, v, 1)$ is the homogeneous numerator specialized at $w = 1$, and

$$\mathrm{ht}(i, j) = bi + aj - ab.$$

The critical triangle consists of integer points with $i < a$, $j < b$, and $\mathrm{ht}(i, j) > 0$. Coefficients retain their individual identities under $w = 1$ by homogeneity.

Theorem 1. *At every interior integer point (i, j) on a Markov sail segment $B_r B_{r+1}$ or $A_{r+1}A_{r+2}$, the coefficient is*

$$\mathcal{M}(i, j) = 4\,\mathrm{ht}(i, j). \tag{1}$$

Consequently both Markov Sail Duality identities hold.

We use the established generalized Cohn-matrix formula [1, Theorem 10.1], with its Christoffel-word realization [2, Sections 2.1.2, 2.7.2 and Theorem 2.7.5]. Put $S = 1 + u + v$ and

$$M = \begin{pmatrix} v & 1 \\ uv & 1+u \end{pmatrix}, \qquad N = \begin{pmatrix} 1+v & 1 \\ uv & u \end{pmatrix}. \tag{2}$$

If C is the product along the lower Christoffel word with b letters M and a letters N, then

$$\mathrm{tr}\, C = SP. \tag{3}$$

Indeed, after $z = 1$, $u = x^2$, $v = y^2$, and conjugation by $\mathrm{diag}(x, 1)$, the source's initial matrices are M and MN. Its modified Christoffel word replaces a horizontal-vertical pair by the latter matrix, giving precisely (3).

Lemma 1. *For $0 < h < a$, the coefficient of the unique critical-triangle monomial of height h in $\mathrm{tr}\, C$ receives a contribution $4h$ from matrix paths with exactly one transition of each type. Every path with $k \ge 1$ transitions of each type has minimum monomial height at least k.*

Proof. Expand the trace as a sum over cyclic states in $\{0, 1\}$, one matrix entry per letter. A $0 \to 1$ entry is 1, and a $1 \to 0$ entry is uv. The diagonal entries have minimum monomials

$$M_{00} = v, \quad M_{11} : 1, \qquad N_{00} : 1, \quad N_{11} = u. \tag{4}$$

Choosing the other summand in a diagonal entry increases height by a or b.

Let $L = a + b$. At the gaps of the Christoffel word, let H be b times the number of preceding N's minus a times the number of preceding M's. Its L values are the integers $-L + 1, \dots, 0$, each once. One direct construction starts at zero and subtracts a if the result stays in this interval, otherwise adds b. These steps give the lower Christoffel path; the values are distinct because each step is $-a$ modulo L and $\gcd(a, L) = 1$. A letter M starts at a height in $[-b + 1, 0]$; a letter N starts in $[-L + 1, -b]$.

Consider a single state-1 arc, beginning with a $0 \to 1$ transition at letter r and ending with a $1 \to 0$ transition at a distinct letter s, in cyclic order. Let m_r, m_s be 1 for an M and 0 for an N. If H_{r-1}, H_{s-1} are the gap heights just before these letters, its minimum monomial has height

$$h = (H_{s-1} - am_s) - (H_{r-1} - bm_r) + a. \tag{5}$$

To check this, if the arc's interior has n letters N and its whole arc, including endpoints, has m letters M, its monomial is $u^{n+1}v^{b-m+1}$. Its height is $b(n+1)-a(m-1)$, which gives (5). The total word has net height zero, so the same calculation holds for an arc crossing its chosen initial gap.

Write $U_s = H_{s-1} - am_s$ and $V_r = H_{r-1} - bm_r$. Their multisets are the unions, with multiplicity, of

$$\begin{aligned} U &: [-L+1, -a] \ \cup \ [-L+1, -b], \\ V &: [-2b+1, -b] \ \cup \ [-L+1, -b]. \end{aligned} \tag{6}$$

In particular, (5) is always at least 1. For $0 < h < a$, it is equivalent to

$$(U_s + L - 1) + (-b - V_r) = h - 1.$$

There are h nonnegative pairs on the left. Every participating value in each multiset has multiplicity two, so there are $4h$ ordered endpoint pairs. None has $r = s$: that choice would give height b for an M or a for an N.

These monomials have both exponents positive. Height $h < a \le b$ therefore forces $i < a$ and $j < b$. For a given h, coprimality gives at most one such pair of exponents, since the difference of two solutions of $bi + aj = ab + h$ is a multiple of $(a, -b)$. Thus the $4h$ paths contribute to the same coefficient.

Finally, a path with k state-1 arcs is obtained from the all-zero state path by making those disjoint local changes. Its minimum height is the sum of the k single-arc heights in (5), each positive. This proves the last assertion too. □

Lemma 2. *An interior point on the lower integer hull above the segment joining $O = (0, b)$ and $T = (a, 0)$, of height $0 < h < a$, receives no contribution from paths with two or more transitions of each type.*

Proof. Such a point Q minimizes a linear function $\ell(i, j) = \alpha i + \beta j$, where $\alpha, \beta \ge 0$ and not both zero, over the integer points of $[0, a] \times [0, b]$ strictly above OT. Write $d = \ell(Q)$. Then

$$\min\{\ell(O), \ell(T)\} < d. \tag{7}$$

For example, if $\alpha a \le \beta b$, the inequality $i/a + j/b > 1$ gives $\ell(i, j) > \alpha a$ when both coefficients are positive. If a coefficient is zero, (7) follows from $i, j > 0$. The other case is symmetric.

Suppose a path with $k \ge 2$ arcs contributes at Q. There can be no nonminimum diagonal choice, since it would add at least a to the already positive height. For each individual state-1 arc, make that change alone to the all-zero path and let Q_r be the resulting exponent point. Then

$$Q = O + \sum_{r=1}^{k} (Q_r - O), \qquad h = \sum_{r=1}^{k} \mathrm{ht}(Q_r). \tag{8}$$

By Lemma 1, every summand height is positive and less than a, and both exponents are positive. Thus each Q_r is itself a critical-triangle integer point, so $\ell(Q_r) \ge d$. If $\ell(O) < d$, applying ℓ to (8) yields $d - \ell(O) \ge k(d - \ell(O))$, a contradiction.

If instead $\ell(T) < d$, make each individual state-0 arc alone to the all-one path. The resulting points R_r likewise have positive heights summing to h and positive exponents: each is a path with one transition of each type. They are therefore in the critical triangle, and

$$Q = T + \sum_{r=1}^{k} (R_r - T).$$

The same argument with T gives a contradiction. Equation (7) ensures that one of these arguments applies. □

Proof of Theorem 1. The source identifies the combined Markov sail with the lower integer hull used in Lemma 2; see [1, pages 20–21]. Write $b/a = [c_1, \dots, c_n]$, with convergents p_k/q_k, $p_{-1} = 0, q_{-1} = 1, p_0 = 1, q_0 = 0$. Its vertices in the numerator's coordinate frame are

$$A_r = (q_{2r-1}, b - p_{2r-1}), \qquad B_r = (a - q_{2r}, p_{2r}). \tag{9}$$

The positive Euclidean remainders $|bq_k - ap_k|$ strictly decrease from a at $k = 0$ to zero at the final convergent. In particular, $\mathrm{ht}(A_1) = b - c_1 a < a$, and $\mathrm{ht}(B_0) = a$. Height is affine on a segment. Consequently every interior critical-triangle point on $B_r B_{r+1}$ or $A_{r+1}A_{r+2}$ has height strictly between 0 and a; the boundary point B_0 is excluded.

Lemmas 1 and 2 show that its coefficient in $\operatorname{tr} C$ is $4h$. Paths with no transitions have monomials divisible by v^b or u^a, so do not contribute. Equation (3) gives the same coefficient in P: the possible contributions from uP and vP would have predecessor heights $h - b < 0$ and $h - a < 0$, outside the established Newton polygon [1, Theorem 3.2]. This proves (1).

For each identity whose opposite vertex is interior, the primitive direction along $B_r B_{r+1}$ is $(-q_{2r+1}, p_{2r+1})$, by the convergent recurrence. Its height increment is $-\mathrm{ht}(A_{r+1})$. Similarly, the primitive direction along $A_{r+1}A_{r+2}$ is $(q_{2r+2}, -p_{2r+2})$, with height increment $-\mathrm{ht}(B_{r+1})$. Applying (1) gives exactly

$$d(B_r B_{r+1}) = -\mathcal{M}(A_{r+1}), \qquad d(A_{r+1}A_{r+2}) = -\mathcal{M}(B_{r+1}).$$

As in the source, $\mathcal{M}$ and differences here concern interior points only. No formula is asserted for the excluded boundary coefficients, and a segment with fewer than two interior integer points supplies no adjacent-difference condition. The Fibonacci cases $a = 1$ have empty critical triangles. □

Task 2. Determine the coefficients on the initial sail segment A_0A_1, including the points with height at least a.

Task 3. Count matrix paths with exactly two transition pairs and obtain explicit formulas for the next layer of coefficients above the sail.

Task 4. Relate the transition-pair decomposition to local operations on weighted snake-graph perfect matchings.

Task 5. Use the path expansion to investigate log-concavity along lines in the full Markov Newton polygon.

References

[1] S. J. Evans, A. P. Veselov, and B. Winn, *Arithmetic and geometry of Markov polynomials*, arXiv:2501.14882v3 (2025); current 28-page journal-hosted manuscript. `https://armj.math.stonybrook.edu/Articles/250708-Winn/article.pdf`.

[2] S. J. Evans, *Quantum numbers and Markov polynomials*, doctoral thesis, Loughborough University, December 2025; deposited February 9, 2026. `https://doi.org/10.26174/thesis.lboro.31250974.v1`.